



Grade 10 Geography

2. Plate Tectonics

Learning Guide



1. What is Plate Tectonics?

- Plate tectonics explains:
 - Movement of **continents**
 - Formation of **earthquakes, volcanoes, and mountains**
- The Earth's crust is:
 - Broken into **large pieces called plates**
 - These plates **float on the semi-molten mantle**



Key Idea: The Earth is NOT solid as one piece — it is **constantly moving**



2. Continental Drift Theory

- Proposed by **Alfred Wegener (1912)**

Main Idea:

- All continents were once joined into one supercontinent called:
 - ☞ **Pangaea**

Pangaea Split Into:

- **Laurasia (North)**
- **Gondwanaland (South)**
- Over millions of years → continents moved to their current positions



3. Movement of Continents

- Continents are **still moving today**
- Speed: **~2 cm per year**

Cause of Movement:

- **Internal forces inside Earth**
- **Convection currents in the mantle**



Important: Movement is slow but continuous



4. Evidence for Continental Drift



1. Jigsaw Fit

- Continents fit like a **puzzle**
- Example:
 - **West Africa fits with East South America**
- More accurate when using **continental shelves**



2. Fossil Evidence

- Same fossils found on different continents

Examples:

- **Mesosaurus** → South America & Africa
- **Glossopteris** → Africa, India, Antarctica, Australia



Conclusion:

These organisms could NOT cross oceans → continents were once joined



3. Matching Rocks & Structures

- Rock layers match across continents

Example:

- Appalachian Mountains (USA) match with Europe

✚ *Conclusion:*

Continents were once part of the same landmass

4. Fold Mountains

- Mountain ranges line up across continents

Example:

- Appalachians (USA) align with Caledonian Mountains (Europe)

✚ *Conclusion:*

Formed together before continents split

5. Past Climate Evidence

- Evidence of climates different from today

Examples:

- Glacial deposits in **Africa & India**
- Coal in **Antarctica**

✚ *Conclusion:*

Continents have moved over time

6. Ocean Floor Spreading (STRONGEST EVIDENCE)

- New crust forms at **mid-ocean ridges**
- Magma rises → cools → pushes plates apart

Evidence:

- Youngest rocks near ridges
- **Magnetic striping patterns** on both sides

✚ *Conclusion:*

Continents are moving apart

○

5. Structure of Plates

- Earth has about **14 major plates**
- Plates are:
 - **Constantly moving**
 - Floating on the **semi-molten mantle**
- Edges of plates = **plate boundaries**

6. What Causes Plate Movement?

- Caused by **convection currents in the mantle**

How it works:

- Hot magma **rises**
- Cool magma **sinks**
- This creates a **circular movement**

✚ *Analogy (page 14):*

Like boiling water moving a pot lid

7. Types of Plate Boundaries

1. Divergent Boundary

- Plates **move apart**

What happens:

- Magma rises
- New crust forms

Forms:

- **Mid-ocean ridges**
- **Volcanic islands (e.g. Hawaii)**

📌 *Key Idea:* New crust is created

2. Convergent Boundary

- Plates **move towards each other**

What happens:

- One plate may **subduct (sink)** under another

Forms:

- **Ocean trenches**
- **Volcanoes**
- **Fold mountains**

📌 *Key Idea:* Crust is destroyed or folded

3. Transform Boundary

- Plates **slide past each other**

What happens:

- No crust created or destroyed
- Causes **earthquakes only**

📌 Example: **San Andreas Fault**

📌 *Key Idea:*

Movement = friction → earthquakes

8. Global Distribution of Activity

- Earthquakes & volcanoes occur along **plate boundaries**

Main Zone:

- 🌐 **Pacific Ring of Fire**

High Activity Areas:

- Western Americas
- Japan
- Indonesia

9. FINAL EXAM SUMMARY

- Earth's crust = **plates that move**
- Movement caused by **mantle convection currents**
- Continents were once **Pangaea**
- Strong evidence:
 - Jigsaw fit
 - Fossils
 - Rock matching
 - Climate evidence
 - Ocean floor spreading
- Plate boundaries:
 - **Divergent** → plates move apart
 - **Convergent** → plates collide
 - **Transform** → plates slide
- Most activity occurs along **plate boundaries (Ring of Fire)**

For the full interactive presentation, visit May Learning Hub:

www.maylearninghub.co.za